Element D. Design Concept Generation, Analysis and Selection

Progress:

Day 1: Meeting Ms. Tatro and deciding what we will design.

Day 2: Creating timeline, team charter, and team portfolio

Day 3: Brainstorming ideas and how to build the book end

Day 4: Finalized the team charter and submitted it

Days 5-10: Drawing and finalizing the design for our bookend

Day 6-8: Finished and submitted component I of the portfolio

Day 11: We changed the materials for the design from popsicle sticks to wood

Day 12: Working on component II of the portfolio finishing element D and creating element E and F.

Redesign Material lists

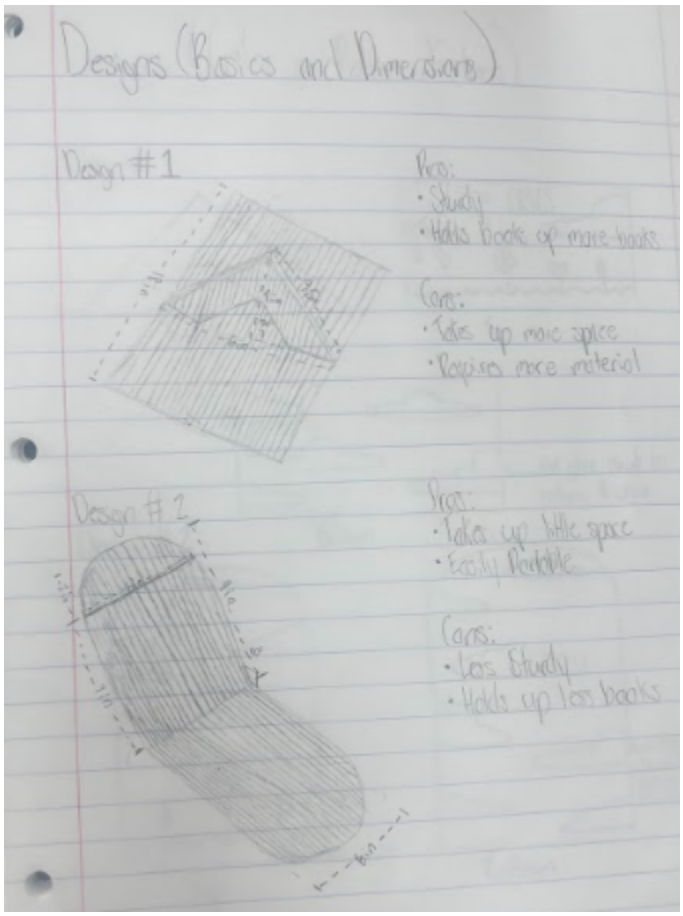
- 1 Thin sheet plastic - 18” x 6” inch
- 1 piece of wood - 12” x 6” inch
- Wood clue
- Wood cutter
- Paint
- Modge podge
- pencil
- Measuring tape
- Hot glue

Original Materials:

- Popsicle sticks
- Metal wire
- Cardboard
- Hotglue
- Paint
- Measuring tape

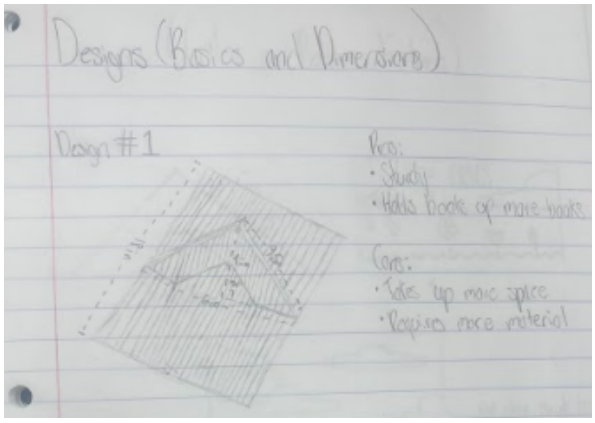
- Scissors
- Woodcutter
- Pencil
- Paper

Prototype sketches

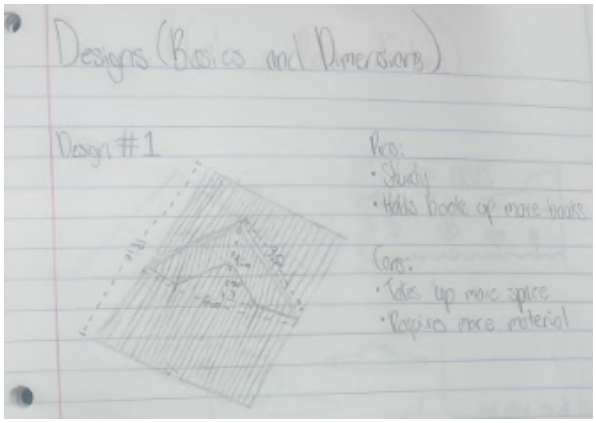


Design Sketch #1

- Idea for sketch design one was created on Mar 26, 2025.
- The base was going to be made out of a thin sheet of one wall cardboard that was then covered with popsicle sticks for reinforcement. The vertical triangular piece was going to be used to hold the book up. To be studier than the base it would use triple-wall cardboard and would be reinforced on both sides again by popsicle sticks.

- 

Improved Design Sketch #1

- Revision for the creation of design was made on April 23rd 2025
- The base of the design has changed in materials as instead of cardboard with metal wiring to hold it and coated in popsicle sticks we switched to a more simple but sturdy plastic for the base. The triangular part of the bookend has been changed to wood so that it is also more sturdy.
- Same design sketch however. 

Design Sketch #2

- Idea for sketch design one was created on Mar 26, 2025.
- The design idea for the second sketch was made as an improvement on the original bookend as the original bookend from the library had a hole in the middle. This design closed the hole making it a rectangle with a semicircle on top of the rectangle for the vertical bookend part. Then for the base we reflected that design onto the base to create the same rectangle with the semi circle on top.

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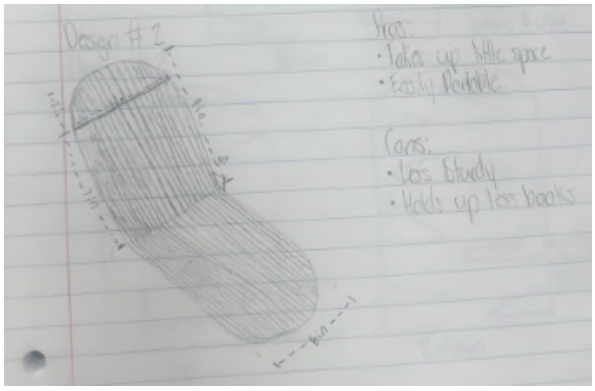
Design #2
Pros:
• Takes up little space
• Easily flexible
Cons:
• Less Sturdy
• Holds up less books